Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

Sentiment-Based Analysis of Online Customer Reviews and Their Influence on Product Ratings

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Introduction

Problem Statement - "A single star rating cannot capture everything — the same rating can reflect completely different situations."

THE PROBLEM -

- Millions of reviews cannot be read manually
- Star ratings are noisy, there is a mismatch between text sentiment & rating is common
- Binary positive/negative labels ignore all five rating tiers

RESEARCH QUESTIONS -

- Are VADER sentiment scores a valid proxy for customer sentiment?
- Do compound scores correlate with star ratings, and how strongly?
- Where does automated classification fail most?

Objectives

RQ1: Can automated VADER sentiment scores meaningfully capture how customers feel, as reflected in their written reviews?

RQ2: Do VADER sentiment scores correlate with the numerical star ratings customers assign, and to what extent?

Methodology

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5-Stage Research Pipeline

1

Dataset Selection

Amazon Fine Food Reviews
10,000 sample (seed=42)

2

Pre-processing

Lowercase, strip HTML, tokenise, remove stop-words

3

Sentiment Scoring

VADER compound score
TextBlob polarity

4

Classifier Evaluation

4-way comparison on 500-review silver set

5

Statistical Analysis

Spearman ρ, Pearson r, Kruskal-Wallis, Mann-Whitney

Methodology

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Dataset & Hypotheses

DATASET -

- Amazon Fine Food Reviews (McAuley & Leskovec, 2013)
- Reviews from 2003–2012 — food products
- Fields: review text, 1–5 star rating, helpfulness votes
- 10,000 reviews sampled (seed = 42)
- Natural class distribution preserved
- Hosted on Stanford SNAP & Kaggle

HYPOTHESIS -

- **H1 :** VADER compound scores are positively & significantly correlated with star ratings (Spearman's $\rho > 0$, $p < 0.001$).
- **H2 :** Sentiment variance differs across rating groups — low-rating reviews show substantially higher dispersion.
- **H3 :** Logistic Regression outperforms VADER on macro F1; VADER outperforms TextBlob.

Tools & Technologies

LANGUAGE -

- Python 3.x

NLP -

- NLTK · VADER (vaderSentiment) · TextBlob

DATA HANDLING -

- Pandas · Numpy

ML -

- Scikit-learn (TF-IDF + Logistic Regression)

STATISTICS -

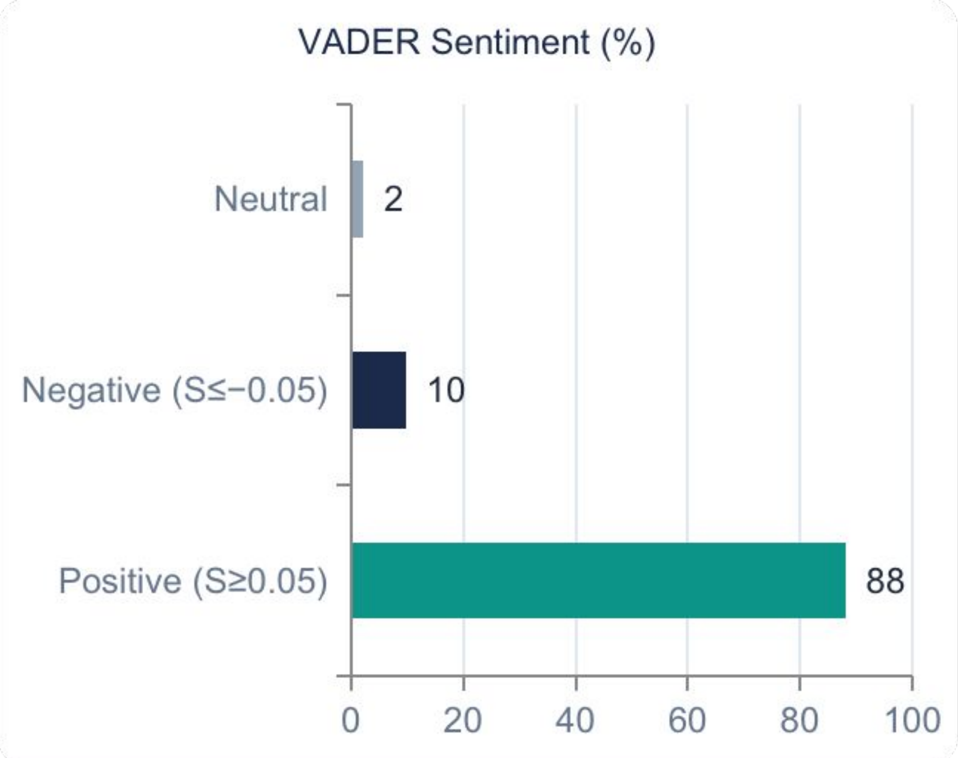
- SciPy (Spearman, Pearson, Kruskal-Wallis, McNemar)

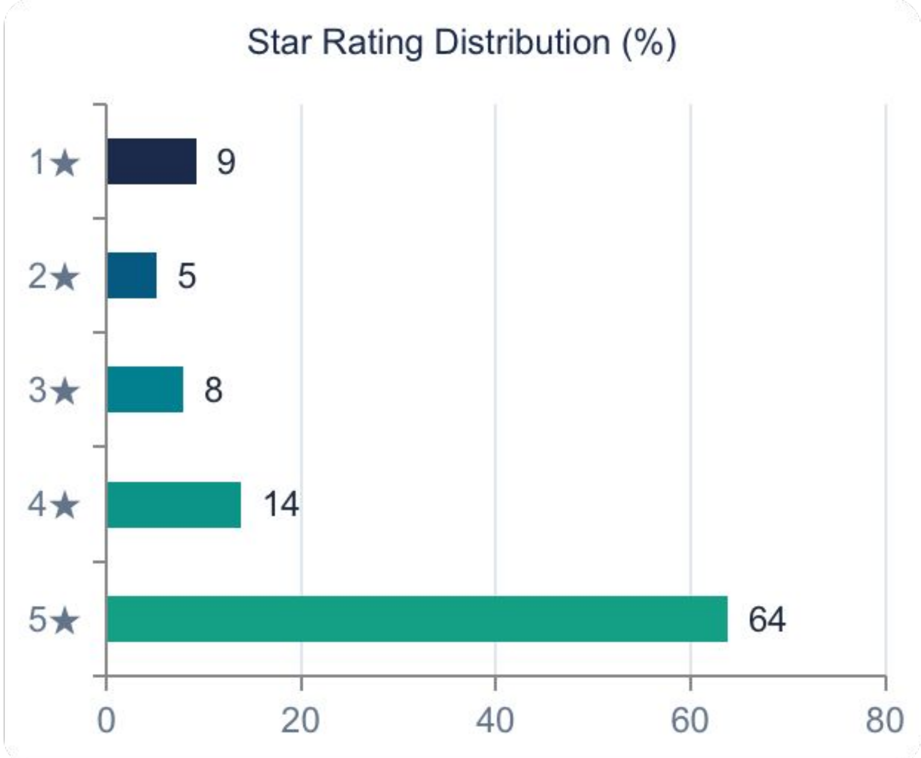
VISUALIZATION -

- Matplotlib · Seaborn

Results

Sentiment & Rating Distribution





Results

H1: Correlation — Sentiment vs Star Rating

0.371	0.504	25.4%
Spearman's ρ (primary)	**Pearson's r (supplementary)**	**Explained Variance (r^2)**
95% CI [0.351, 0.389] $p < 0.001$	95% CI [0.484, 0.524] $p < 0.001$	Cohen's d: Medium effect

H1 SUPPORTED Real & meaningful correlation — but 75% of rating variance lies outside the review text (delivery, price, brand expectations).

Results: Correlation

H3: Classifier Comparison

System	Accuracy	Macro F1	Neg. Recall	Training Data
Majority-Class Baseline	50.0%	0.333	0.00 ✗	None
TextBlob	84.2%	0.833	0.61	None
VADER *	100.0% *	1.000 *	1.00 *	None
Logistic Regression	84.0%	0.840	0.79 ✓	400 labelled

•Logistic Regression (84% F1) is the most generalisable result — trained on 400 domain-specific examples

•Negative recall is the hardest across all systems — negative reviews are linguistically noisier

•LR outperforms TextBlob on negative recall: 0.79 vs 0.61 — showing the value of domain labels

Conclusion

- Spearman ρ used as the primary measure (not Pearson-only), giving a more conservative and accurate estimate of association between ordinal ratings and continuous scores.
- Quantified sentiment dispersion across all five rating levels — revealing that negative labels (1★) are substantially noisier than positive ones (5★), a critical insight for classifier design.
- Four-way classifier comparison includes a majority-class baseline and explicitly accounts for the evaluation bias introduced by silver-standard label construction that was rarely done in prior work.

Conclusion

- VADER correlates with star ratings ($\rho = 0.371$, $r = 0.504$) — real but moderate. Only 25.4% of rating variance is explained by review text.
- 1★ reviews average near-neutral sentiment ($\mu = 0.044$, $\sigma = 0.637$) — dissatisfied customers write inconsistently. Negative labels are inherently noisier.
- Negative recall is the weakest point across all systems, a persistent challenge driven by the variance of 1★ review language.
- Star ratings and VADER sentiment scores are related but not interchangeable, using one as a ground truth for the other introduces systematic noise.

Future Scope

- Annotate a subset including hard cases (sarcasm, irony, mixed sentiment) for a proper benchmark.
- Fine-tune contextual models to quantify accuracy–efficiency trade-offs and test the 1★ variance problem.
- Decompose reviews by attribute (taste, delivery, packaging) to explain rating vs text divergences.
- Address 88.1% positive-class dominance to improve negative and neutral recall in production.
- Test on electronics, healthcare, hospitality reviews to assess generalisability.

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Thank You